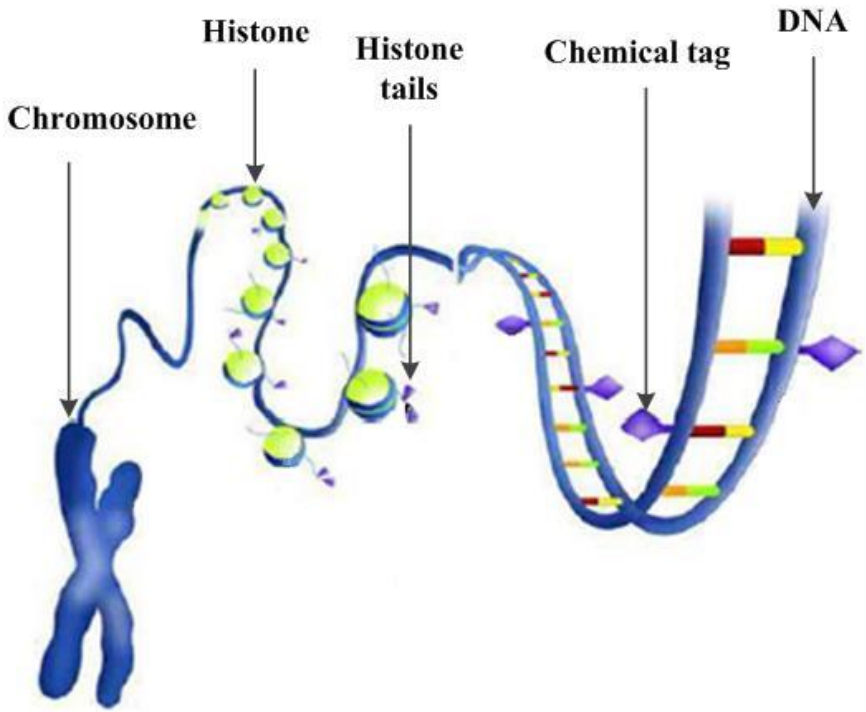
SCSJ 3563
COMPUTATIONAL INTELLIGENCE

Dr. Shafaatunnur Hasan
Prof. Dr. Siti Mariyam Shamsuddin
Universiti Teknologi Malaysia

Email: shafaatunnur@utm.my
/shafaatunnur@gmail.com

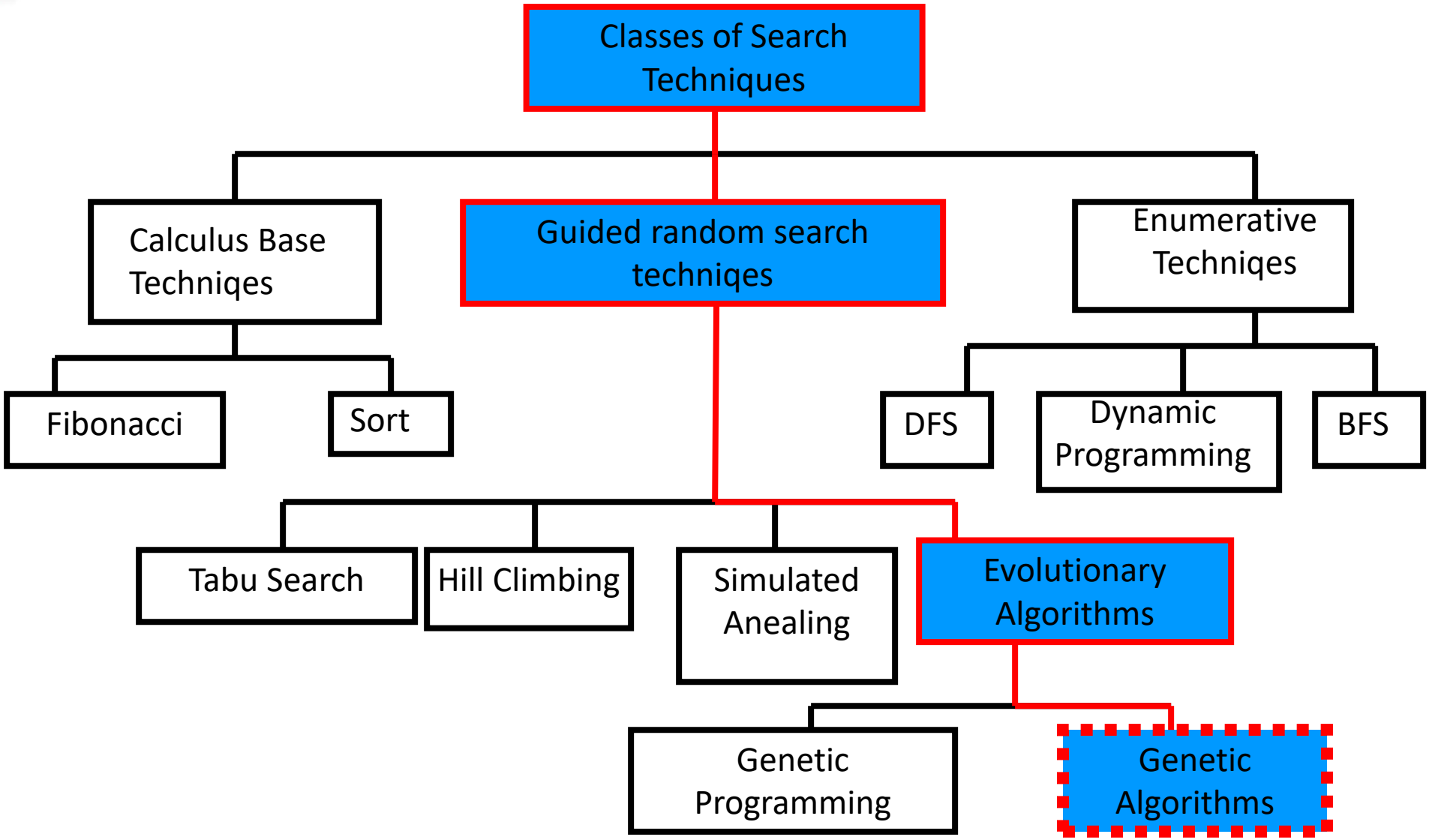


GENETIC ALGORITHM (GA)



A1	0	0	0	0	0	0	Gene
A2	1	1	1	1	1	1	Chromosome
A3	1	0	1	0	1	1	
A4	1	1	0	1	1	0	Population

Genetic Algorithms (GA) Overview



Genetic Algorithms (GA) Overview

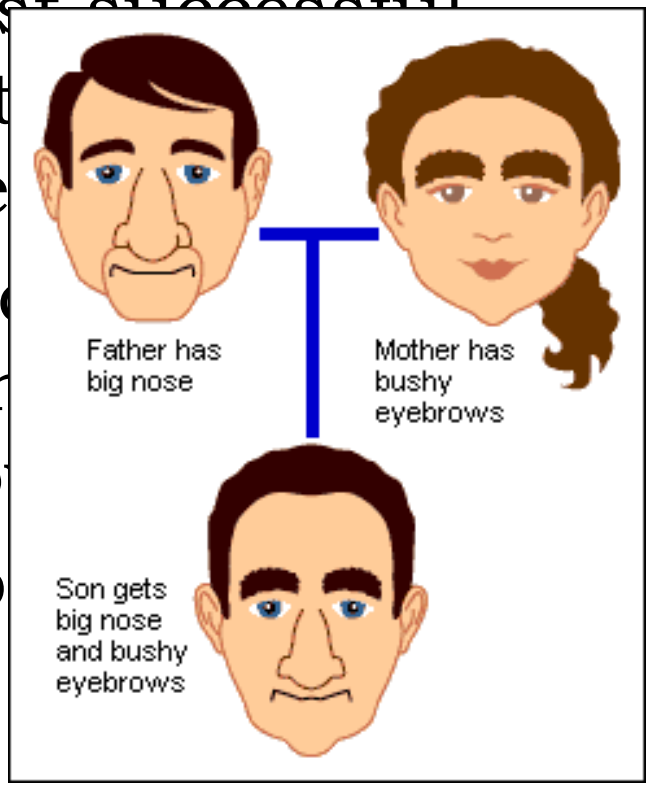
- A class of heuristic optimization algorithms
- Inspired by the biological evolution process.Uses concepts of “Natural Selection” and “Genetic Inheritance” (Darwin 1859)
- GA is developed by John Holland (1975)
- Particularly well suited for hard problems where little is known about the underlying search space. Widely-used in business, science and engineering

Genetic Algorithms (GA) Overview

A genetic algorithm maintains a population of candidate solutions for the problem at hand, and makes it evolve by iteratively applying a set of stochastic operators

What is Stochastic operators

- ***Selection*** replicates the mos
  solutions found in a populat
  proportional to their relative
- ***Recombination/Crossover***
  two distinct solutions and th
  mixes their parts to form no
- ***Mutation*** randomly perturb
  solution



GA PROCEDURE

produce an initial population of individuals

evaluate the fitness of all individuals

while termination condition not met **do**

    select fitter individuals for reproduction

    recombine/crossover between individuals

    mutate individuals

    evaluate the fitness of the modified individuals

    generate a new population

End while

Step by Step GA for Function Optimization Problem

Step by Step Calculation

Problem:

Maximize the function of $f(x) = x^2$, where x varies between 1 and 31.

Solutions:

1. Code the decision variables, x of the problem as finite length string. Code it as binary unsigned integer of length 5. Select an initial population of size 4.

Step by Step Calculation

Calculate the decision variable, x

String	Binary encoding	Decision variables, x
S_1	01100_2	?
S_2	11001_2	?
S_3	00101_2	?
S_4	10011_2	?

Step by Step Calculation

For string S_1 = 01100, the decision variables, x value is:

$01100=0*2^0+0*2^1+1*2^2+1*2^3+0*2^4=0+0+4+8+0=12$

String	Binary encoding	Decision variables, x
S_1	01100_2	12
S_2	11001_2	25
S_3	00101_2	5
S_4	10011_2	19

$00101=1*2^0+0*2^1+1*2^2+0*2^3+0*2^4=1+0+4+0+0=5$

For string S_4 = 10011, the decision variables, x value is:

$10011=1*2^0+1*2^1+0*2^2+0*2^3+1*2^4=1+2+0+0+16=19$

Step by Step Calculation

2. Find the fitness function, $f(x) = x^2$.

String	Binary encoding	Decision variables, x	fitness function $f(x) = x^2$
S_1	01100_2	12	?
S_2	11001_2	25	?
S_3	00101_2	5	?
S_4	10011_2	19	?
Sum			?
Mean			?
Max			?

Step by Step Calculation

String	Binary encoding	Decision variables, x	fitness function $f(x) = x^2$
S_1	01100_2	12	144
S_2	11001_2	25	625
S_3	00101_2	5	25
S_4	10011_2	19	361
Sum			**1155**
Mean			**288.75**
Max			**625**

Sum of the fitness value=$\sum f(x)$ =144+625+25+361=1155

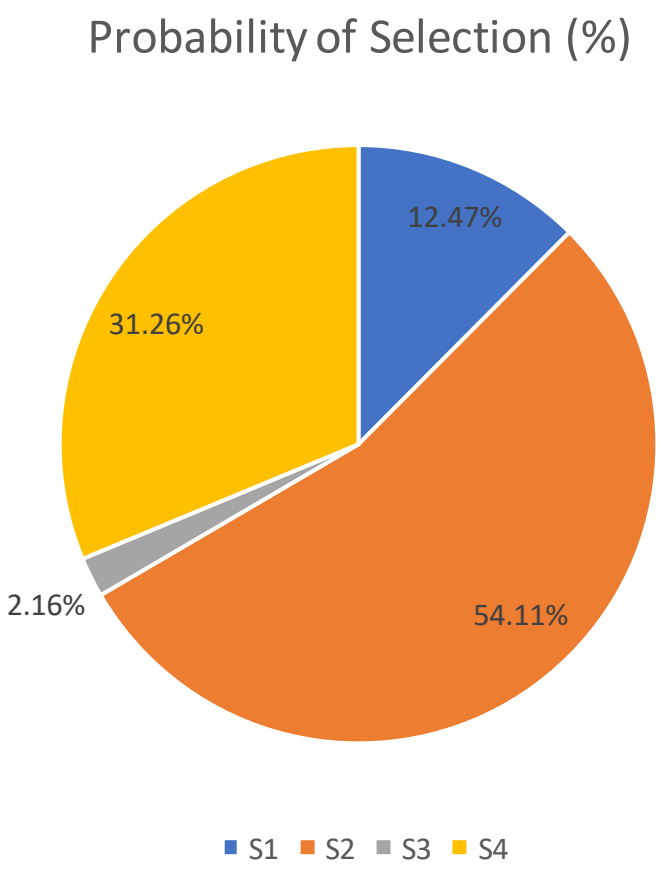
Step by Step Calculation

Then, calculate the probability of selection, expected and actual count (Roulette Wheel) for the function.

String	Binary encoding	Decision variables, x	fitness function $f(x) = x^2$	Probability of selection, $P_i = f(x) / \sum f(x)$
S_1	01100_2	12	144	?
S_2	11001_2	25	625	?
S_3	00101_2	5	25	?
S_4	10011_2	19	361	?
Sum			1155	?
Mean			288.75	?
Max			625	?

Step by Step Calculation

The Probability of selection, $P_i = f(x)/\sum f(x)$ **:**

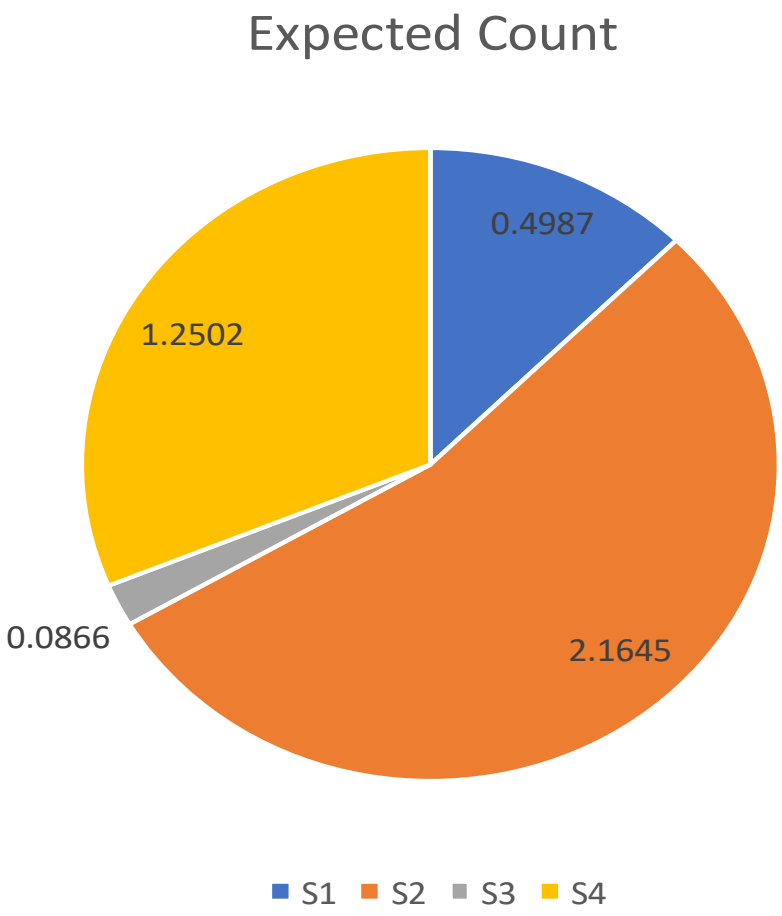


Step by Step Calculation

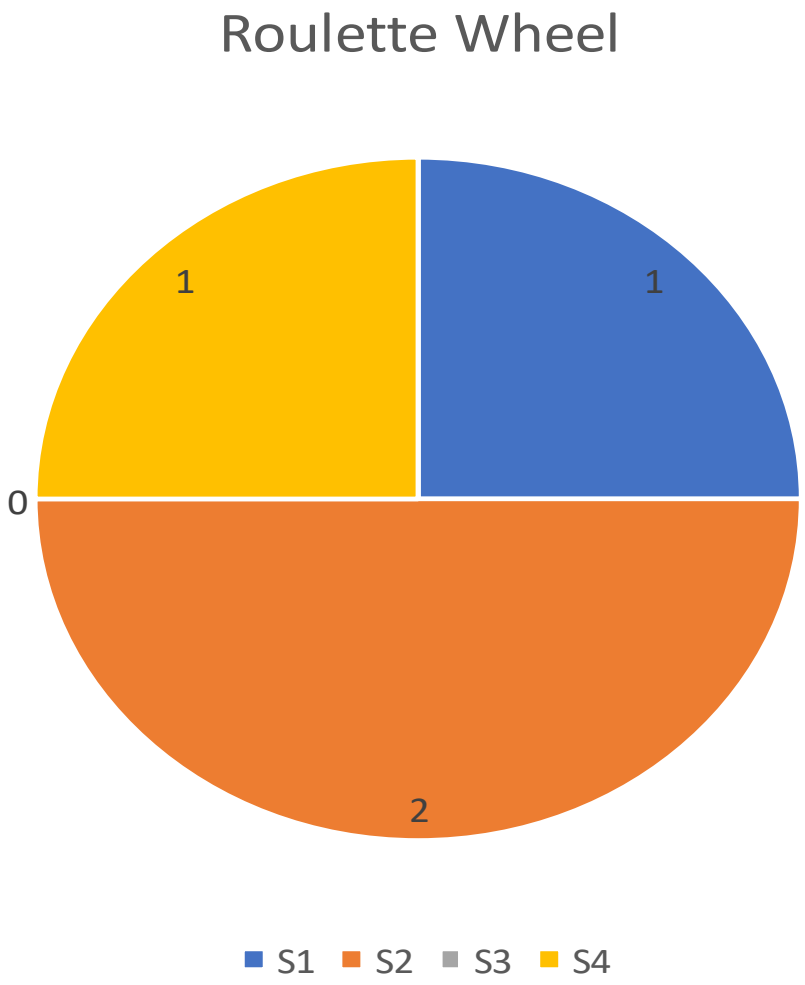
The Probability of selection, $P_i = f(x)/\sum f(x)$

String	Binary encoding	Decision variables, x	fitness function $f(x) = x^2$	Probability of selection, $P_i = f(x)/\sum f(x)$
S_1	01100_2	12	144	0.1247
S_2	11001_2	25	625	0.5411
S_3	00101_2	5	25	0.0216
S_4	10011_2	19	361	0.3126
Sum			1155	1.000
Mean			288.75	0.2500
Max			625	0.5411

Step by Step Calculation



Step by Step Calculation



Step by Step Calculation

3. Apply Recombination/Crossover operator. A single-point crossover is performed and new offspring (children) is produced.

String	Mating pool	Crossover point	Offspring after crossover
S_1	01100	3	01101
S_2	11001	3	11000
S_3	11001	2	11011
S_4	10011	2	10001

Step by Step Calculation

Re-calculate x values and evaluate new fitness after Crossover.

Table 2. Crossover

String	Offspring before crossover	Offspring after crossover	Decision variables, x'	fitness Function, $f(x') = x^2$
S_1	01100	01101	13	169
S_2	11001	11000	24	576
S_3	11001	11011	27	729
S_4	10011	10001	17	289
Sum				1763
Mean				440.75
Max				729

Step by Step Calculation

4. Apply Mutation operator

String	Offspring (children) after crossover	Mutation Chromosome for flipping	Offspring (children) after mutation
S_1	01101	10000	11101
S_2	11000	00000	11000
S_3	11011	00000	11011
S_4	10001	00100	10101

Step by Step Calculation

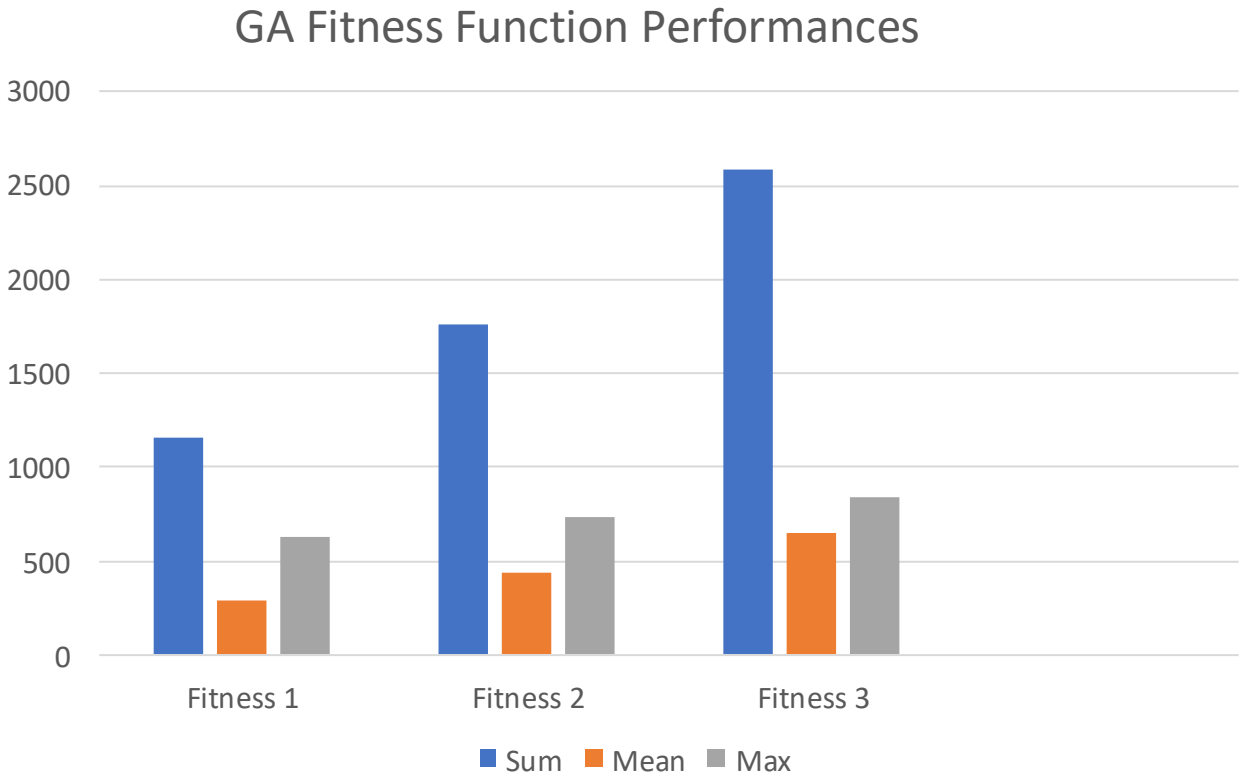
Re-calculate x values and evaluate new fitness after Mutation.

Table 3. Mutation

String	Offspring (children) after crossover	Mutation Chromosome for flipping	Offspring (children) after mutation	Decision variables, x''	fitness Function, $f(x'') = x^2$
S_1	01101	10000	11101	29	841
S_2	11000	00000	11000	24	576
S_3	11011	00000	11011	27	729
S_4	10001	00100	10101	21	441
Sum					2587
Mean					646.75
Max					841

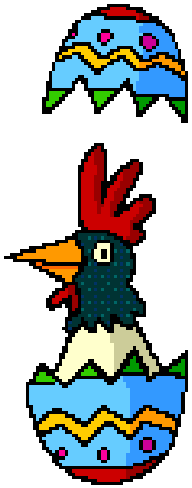
Step by Step Calculation

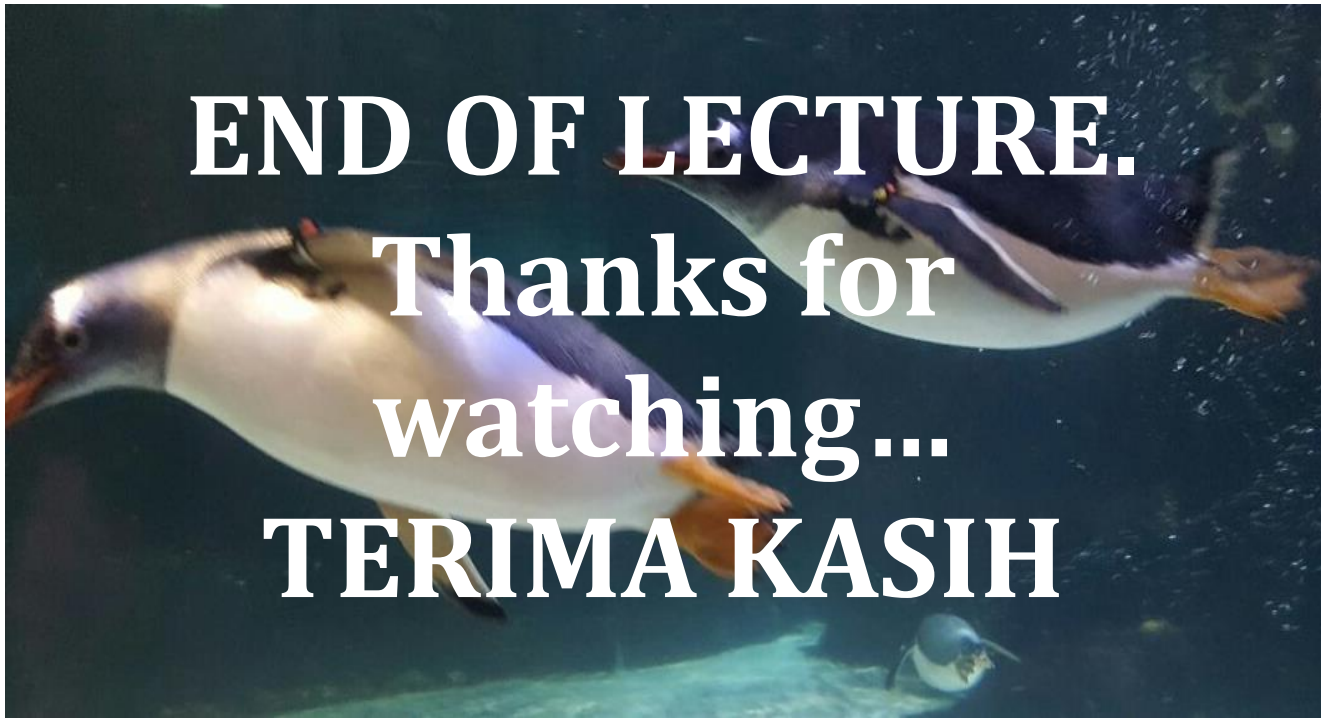
5. Stop the cycle generation and evaluate GA performances in-terms of sum, mean and maximum fitness.



Conclusion

- Referring to Table-1, Table-2, Table-3, we can conclude that the total sum, maximum and the average fitness performances have been improved significantly. In one generation, the average fitness has improved from 288.75 to 646.75, with 124%. While, the maximum fitness has improved 625 to 841, with 35%. In addition, the total fitness has been increased from 1155 to 2587 with 124% in a single generation.
- The best string from initial population (randomly selected) 01100,11001,00101,10011 is 11001 (see Table 1). Thus, after crossover and mutation (see Table-2&-3), new Offspring (11101) has been produced which is a better choice.




END OF LECTURE.
Thanks for watching…
TERIMA KASIH